

GPU-Vectorized DPM $S_{26}^{(3)}$ Spectral Atlas Engine

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Abstract

We present a GPU-accelerated Dipole Moment (DPM) spectral atlas engine that computes the full $S_{26}^{(3)}$ spectral profile across all 26 quantum states using batched matrix multiplication. The engine supports PyTorch (CUDA/CPU), NumPy, and pure-Python backends with automatic fallback, achieving sub-3 ms atlas generation on modern hardware. ALMA Cycle 12 target profiles for star-forming regions are generated natively.

§1 DPM Spectral Formulation

The DPM spectral atlas at angular frequency ω is:

$$\text{DPM}(\omega) = \sum_{i=1}^{26} c_i \cdot \Phi_i(\omega)$$

where the layer coefficients are:

$$c_i = \frac{[\text{SSq}]^i}{i^{26}} \cdot R_n(i, 3)$$

and the per-layer Gaussian profile with slight broadening is:

$$\Phi_i(\omega) = \exp\left(-\frac{(\omega - \omega_{\text{SCM}})^2}{2\sigma_i^2}\right), \quad \sigma_i = \sigma_G(1 + 0.02(i - 1))$$

§2 Matmul Formulation

The atlas computation is formulated as a single matrix multiplication:

$$\mathbf{A} = \mathbf{C} \cdot \mathbf{G}$$

where: - $\mathbf{C} = [c_1, \dots, c_{26}]$ has shape $(26,)$ - $\mathbf{G}_{ij} = \Phi_i(\omega_j)$ has shape $(26, N_{\text{freq}})$ - $\mathbf{A}$ has shape $(N_{\text{freq}},)$

This formulation is optimal for GPU execution via `torch.matmul`.

Backend	Dependency	Device	Typical Speed
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§3 Backend Architecture

Backend	Dependency	Device	Typical Speed
PyTorch	torch	CUDA/CPU	<3 ms (512 bins)
NumPy	numpy	CPU	~5 ms
Pure Python	None	CPU	~50 ms

Automatic detection selects the best available backend at runtime.

§4 ALMA Cycle 12 Integration

The atlas engine generates predicted DPM profiles at ALMA observing bands:

Band	Frequency Range	Application
B3	84–116 GHz	Continuum
B4	125–163 GHz	Intermediate
B6	211–275 GHz	CO, HCN, CS
B7	275–373 GHz	High-frequency molecular
Full	84–950 GHz	Complete atlas

Profiles are gravity-scaled per target: $A_k(\omega) = (g_k/g_\odot) \cdot \text{DPM}(\omega)$

§5 Calibration Table

Parameter	Value	Source
$S_{26}^{(3)}$	9.50×10^{-2}	Ramanujan acceleration
ω_{SCm}	7.854×10^{12} rad/s	SCm phonon resonance
σ_{G}	5.027×10^{11} rad/s	UQFF linewidth
Peak DPM	9.498×10^{-2}	Sum of c_i at center
Peak freq	1.2485 THz	Atlas maximum
N_layers	26	Quantum states

§6 Line Identification

DPM spectral lines are identified via local maxima detection above a threshold fraction of the global peak. At standard parameters, the atlas produces a single dominant line at 1.25 THz with FWHM governed by σ_{G} .

§7 SM Gate Compliance

- **G1 (Theoretical Foundation):** DPM derived from $S_{26}^{(3)}$ Ramanujan polylogarithm
- **G2 (Mathematical Consistency):** Matmul formulation preserves summation accuracy
- **G3 (Numerical Stability):** Float64 precision, graceful backend fallback
- **G4 (Physical Motivation):** SCm phonon resonance at 1.25 THz
- **G5 (Observational Bridge):** ALMA band profile generation
- **G6 (Reproducibility):** Deterministic, backend-independent results

References

- `source10_gpu_dpm_atlas.py`: Implementation (11/11 tests pass)
- `production_scaling_v17.py`: Kernels `kernel_gpu_dpm_atlas_peak`, `kernel_dpm_line_fwhm`
- PAPER_877: Three-Assumption UQFF Cosmogenesis